Anand Bhattad

Pixels · Perception · Physics

Assistant Professor of Computer Science · DSAI & Institute of Robotics
Johns Hopkins University

bhattad@jhu.edu anandbhattad.github.io Baltimore, MD

Research

Generative models look right without being right. My 3P Vision group builds the tools to tell the difference — to read out what they know and don't, and how to improve them — along three Ps:

Pixels — controllable generative models for relighting, compositing, and editing.

Perception — reading out the scene structure models implicitly know: intrinsics, geometry, and parts.

Physics — testing and enforcing physical and projective consistency, and carrying it to robots.

The group looks beyond generative models too: vision architectures, 3D perception, multisensory perception for embodied AI, and generative models for computational mechanics.

Keywords: Visual Generative Models; 3D Vision; Vision-based Robotics; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Physics Awareness in Large Vision Models.

Education

University of Illinois Urbana-Champaign (UIUC), USA

Ph.D., Computer Science 2019 – 2024

Thesis: **Exploring Knowledge in Generative Models**

Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018

M.S., Civil Engineering 2015 – 2017

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering 2011 – 2015

Work Experience

Johns Hopkins University, Assistant Professor of Computer Science 2025 – present
Toyota Technological Institute at Chicago, Research Assistant Professor 2023 – 2025
University of California Berkeley, Visiting Scholar; Host: Alexei A. Efros 2024
Allen Institute for AI (Ai2), Research Intern with Aniruddha Kembhavi 2023
Intel, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021
NVIDIA, Research Intern with Andrew Tao and Bryan Catanzaro 2020
Fyusion Inc, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019
Siemens Corporate Technology, Computer Vision & DL Intern with Jan Ernst 2017
Michigan State University, Research Intern with Nizar Lajnef 2014
IIIT Hyderabad, Research Intern with Ramacharla Pradeep Kumar 2013

Selected Honors and Awards

Best Paper Award in Robot Learning, ICRA 2026
Outstanding Reviewer, ICCV 2023
Best Paper Finalist, CVPR 2022
Outstanding Emergency Reviewer, CVPR 2021
Excellent Teaching Assistant, Intro Computing: Engrg & Sci 2016

Selected Publications

Self-selected publications (not necessarily the most cited work).

- OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing
  on arXiv
- Generative Cinematographer: Composing Camera and Object Motion in 3D
  on arXiv
- PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams
  on arXiv
- Shadows Don't Lie and Lines Can't Bend! Generative Models Don't Know Projective Geometry...For Now
  CVPR 2024
- StyleGAN Knows Normals, Depth, Albedo, and More
  NeurIPS 2023
- PRINCIPIA: Relational Physics Tests for Video Models
  NeurIPS 2026 (Evaluations & Datasets Track)
- Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting
  NeurIPS 2025
- Latent Intrinsics Emerge from Training to Relight
  NeurIPS 2024 (**Spotlight**)
- Generative Blocks World: Moving Things Around in Pictures
  ICLR 2026
- Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning
  ICRA 2026 (**Best Paper Award in Robot Learning**)
- LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting
  CVPR 2025

Papers Under Review

⋆ denotes equal contribution, † indicates equal advising, and underlined authors are students I advised or co-mentored

6. **OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing**
   Ariel Slepyan, Geeta Chandra Raju Bethala, Jiafeng Gu, Anagha Rajendra Dangle, Yinzhe Zhou, Hongyuan Fu, Jade Sim, Tuhin Basu, Xinyi Yin, Md Faizal Karim, Gregory Hager†, Homanga Bharadhwaj†, **Anand Bhattad**†, Krishna Murthy Jatavallabhula†, Nitish V. Thakor†
   on arXiv
   Paper Project Code
5. **OmniRay: Camera-Agnostic Monocular 3D Reconstruction with Composite Ray Fields**
   Namitha Guruprasad, Abhay Yadav, Cheng Peng, Rama Chellappa†, **Anand Bhattad**†
   on arXiv
   Paper Project

4. **Generative Cinematographer: Composing Camera and Object Motion in 3D**
   Jiahan Zhang, Chaohao Yang, Namitha Guruprasad, Vivekjyoti Banerjee, Trong-Tung Nguyen, Alan Yuille, **Anand Bhattad**
   on arXiv
   Paper Project Code
3. **PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams**
   Trong-Tung Nguyen, **Anand Bhattad**
   on arXiv
   Paper Project Code
2. **FracGen: Learning How Objects Stretch and Tear with Physics-Informed Video Generation**
   Trong-Tung Nguyen, Jiahan Zhang, **Anand Bhattad**
   on arXiv
   Paper Project
1. **ViT-5: Vision Transformers for The Mid-2020s**
   Feng Wang, Sucheng Ren, Tiezheng Zhang, Predrag Neskovic, **Anand Bhattad**, Cihang Xie, Alan Yuille
   on arXiv
   Paper Code

Refereed Conference Papers

32. **PRINCIPIA: Relational Physics Tests for Video Models**
    Varun Varma Thozhiyoor*, Shivam Tripathi*, Venkatesh Babu Radhakrishnan, **Anand Bhattad**
    NeurIPS 2026 (Evaluations & Datasets Track)
    Paper Project Code
31. **Thinking in Boxes: 3D Editing in Real Images Made Easy**
    Pradhaan S Bhat*, Naveen Chandra*, Rishubh Parihar, Vaibhav Vavilala, Venkatesh Babu, D.A. Forsyth, **Anand Bhattad**
    NeurIPS 2026
    Paper Project Code
30. **SyncLight: Single-Edit Multi-View Relighting**
    David Serrano-Lozano, **Anand Bhattad**†, Luis Herranz, Jean-François Lalonde†, Javier Vazquez-Corral†
    NeurIPS 2026
    Paper Project Code
29. **RATS! Patches Talk Through Registers: Emergent Parts in Register Attention Transformers**
    Timing Yang, Predrag Neskovic, Jansen Seheult, Wenchao Han, **Anand Bhattad**, Alan Yuille, Feng Wang
    NeurIPS 2026
    Paper Code
28. **Context-Aware Autoregressive Image Generation for Emerging Reasoning Properties**
    Jixuan Ying, Haoyu Liu, Timing Yang, Tingyu Zhu, Predrag Neskovic, **Anand Bhattad**, Cihang Xie, Zeyu Zheng, Alan Yuille, Feng Wang
    NeurIPS 2026
27. **SyncFix: Fixing 3D Reconstructions via Multi-View Synchronization**
    Deming Li, Abhay Yadav, Cheng Peng, Rama Chellappa, **Anand Bhattad**
    ECCV 2026
    Paper Project Code

26. **Relighting as a Probe of Visual Priors via Augmented Latent Intrinsics**
Xiaoyan Xing, Xiao Zhang, Sezer Karaoglu, Theo Gevers, **Anand Bhattad**
ICML 2026
Paper Project Code

25. **Objects in Generated Videos Are Slower Than They Appear: Models Suffer Sub-Earth Gravity and Don't Know Galileo's Principle...for now**
Varun Varma Thozhiyoor, Shivam Tripathi, Venkatesh Babu Radhakrishnan, **Anand Bhattad**
CVPR 2026 (Findings)
Paper Project Code

24. **Name That Part: 3D Part Segmentation and Naming**
Soumava Paul, Prakhar Kaushik, Ankit Vaidya, **Anand Bhattad**, Alan Yuille
CVPR 2026 (Findings)
Paper Project Code

23. **Generalizable Sparse-View 3D Reconstruction from Unconstrained Images**
Vinayak Gupta, Chih-Hao Lin, Shenlong Wang, **Anand Bhattad**, Jia-Bin Huang
CVPR 2026
Paper Project Code

22. **Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning**
Tianchong Jiang, Jingtian Ji, Xiangshan Tan, Jiading Fang, **Anand Bhattad**, Vitor Guizilini, Matthew R. Walter
ICRA 2026 (**Oral**; **Best Paper Award in Robot Learning**)
Paper Project Code

21. **Generative Blocks World: Moving Things Around in Pictures**
Vaibhav Vavilala, Seemandhar Jain, Rahul Vasanth, D.A. Forsyth, **Anand Bhattad**
ICLR 2026
Paper Project Code

20. **SpotLight: Shadow-Guided Object Relighting via Diffusion**
Frédéric Fortier-Chouinard, Zitian Zhang, Louis-Etienne Messier, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
3DV 2026
Paper Project Code

19. **Improved Convex Decomposition with Ensembling and Boolean Primitives**
Vaibhav Vavilala, Florian Kluger, Seemandhar Jain, Bodo Rosenhahn, **Anand Bhattad**, D.A. Forsyth
3DV 2026
Paper

18. **Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting**
Anand Bhattad, Konpat Preechakul, Alexei A. Efros
NeurIPS 2025
Paper Project

17. **LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting**
Xiaoyan Xing, Konrad Groh, Sezer Karaoglu, Theo Gevers, **Anand Bhattad**
CVPR 2025
Paper Project Code

16. **ScribbleLight: Single Image Indoor Relighting with Scribbles**
Jun Myeong Choi, Annie N. Wang, Pieter Peers, **Anand Bhattad**, Roni Sengupta
CVPR 2025
Paper Project Code

15. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
Zhi-Hao Lin, Bohan Liu, Yi-Ting Chen, Kuan-Sheng Chen, D.A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong Wang
3DV 2025
Paper Project Code

14. **ZeroComp: Zero-shot Object Compositing from Image Intrinsics via Diffusion**
Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
WACV 2025 (**Oral presentation**)
Paper Project Code

13. **Latent Intrinsics Emerge from Training to Relight**
Xiao Zhang, William Gao, Seemandhar Jain, Michael Maire, D.A. Forsyth, **Anand Bhattad**
NeurIPS 2024 (**Spotlight presentation**)
Paper Project Code

12. **From an Image to a Scene: Learning to Imagine the World from a Million 360° Videos**
Matthew Wallingford, **Anand Bhattad**, Aditya Kusupati, Vivek Ramanujan, Matt Deitke, Sham Kakade, Aniruddha Kembhavi, Roozbeh Mottaghi, Wei-Chiu Ma, Ali Farhadi
NeurIPS 2024
Paper Project Code

11. **Videoshop: Localized Semantic Video Editing with Noise-Extrapolated Diffusion Inversion**
Xiang Fan, **Anand Bhattad**†, Ranjay Krishna†
ECCV 2024
Paper Project Code

10. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models Don't Know Projective Geometry...For Now**
Ayush Sarkar*, Hanlin Mai*, Amitabh Mahapatra*, Svetlana Lazebnik, D.A. Forsyth, **Anand Bhattad**
CVPR 2024
Paper Project Code

9. **StyLitGAN: Image-based Relighting via Latent Control**
Anand Bhattad, James Soole, D.A. Forsyth
CVPR 2024
Paper Project Code

8. **StyleGAN Knows Normals, Depth, Albedo, and More**
Anand Bhattad, Daniel McKee, Derek Hoiem, D.A. Forsyth
NeurIPS 2023
Paper

7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
Oscar Michel, **Anand Bhattad**, Eli Vanderbilt, Ranjay Krishna, Aniruddha Kembhavi, Tanmay Gupta
NeurIPS 2023
Paper Project Code

6. **Improving Equivariance in State-of-the-Art Supervised Depth and Normal Predictors**
Yuanyi Zhong, **Anand Bhattad**, Yuxiong Wang, D.A. Forsyth
ICCV 2023
Paper Code

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
Anand Bhattad, D.A. Forsyth
3DV 2022 (**Spotlight presentation**)
Paper Project Code

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering**
   Liwen Wu, Jae Yong Lee, **Anand Bhattad**, Yuxiong Wang, D.A. Forsyth
   CVPR 2022 (**Best Paper Finalist**; **Oral presentation**)
   Paper Project Code
3. **View Generalization for Single Image Textured 3D Models**
   Anand Bhattad, Aysegul Dundar, Guilin Liu, Andrew Tao, Bryan Catanzaro
   CVPR 2021
   Paper Project
2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
   Anand Bhattad*, Min-Jin Chong*, Kaizhao Liang, Bo Li, D.A. Forsyth
   ICLR 2020
   Paper Code
1. **Improved Style Transfer with Calibrated Metrics**
   Mao-Chuang Yeh*, Shuai Tang*, **Anand Bhattad**, Chuhang Zou, D.A. Forsyth
   WACV 2020
   Paper

Refereed Workshop Papers

5. **Intrinsic LoRA: A Generalist Approach for Discovering Knowledge in Generative Models**
   Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
   CVPR-W 2024 (**Oral presentation**)
   workshops: Generative Models for CV, AI for 3D Gen, SyntaGen, SynthData4CV, Dataset Distillation
   Paper Project Code
4. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
   Zhi-Hao Lin, Bohan Liu, Yi-Ting. Chen, D.A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong. Wang
   CVPR-W 2024
   workshop: SynthData4CV
   Paper Project Code
3. **MIMIC: Masked Image Modeling with Image Correspondences**
   Kalyani Marathe, Mahtab Bigverdi, Nishat Khan, Tuhin Kundu, Patrick Howe, Sharan Ranjit S, **Anand Bhattad**, Aniruddha Kembhavi, Linda G. Shapiro, Ranjay Krishna
   CVPR-W 2024
   workshop: 3D with Multi-View Supervision (Archival long paper)
   Paper Code
2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   Anand Bhattad*, Min-Jin Chong*, Kaizhao Liang, Bo Li, D.A. Forsyth
   CVPR-W 2019 (**Oral presentation**)
   workshop: Adversarial Machine Learning in Real-World Computer Vision Systems
   Paper Code
1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   Anand Bhattad, Jason Rock, D.A. Forsyth
   CVPR-W 2018 (**Oral presentation**)
   workshop: Vision with Biased and Scarce Data
   Paper

Technical Reports

4. **Generative Models: What Do They Know? Do They Know Things? Let's Find Out!**
   Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
   Technical Report, arXiv 2024
   Paper Project Code

3. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
   Vaibhav Vavilala, Seemandhar Jain, Rahul Vasanth, **Anand Bhattad**, D.A. Forsyth
   Technical Report, arXiv 2023
   Paper

2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
   Anand Bhattad, Viraj Shah, Derek Hoiem, D.A. Forsyth
   Technical Report, arXiv 2023
   Paper Project

1. **SIRfyN: Single Image Relighting from your Neighbors**
   D.A. Forsyth, **Anand Bhattad**, Pranav Asthana, Yuanyi Zhong, Yuxiong Wang
   Technical Report, arXiv 2021
   Paper

Patents

1. **Intrinsic-ControlNet: Zero-shot Object Compositing from Image Intrinsics**
   Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
   Pending

Advising

Postdoctoral Researchers

- **Jamila Taaki** (JHU, primary advisor Laurent Pueyo) 2026 – now

Doctoral Students

- **Deming (Remus) Li** (JHU, co-advised with Rama Chellappa) 2026 – now
- **Jiahan Zhang** (JHU) 2026 – now
- **Trong-Tung Nguyen** (JHU) 2026 – now

Master's Students

- **Chaohao Yang** (JHU) 2026 – now
- **Joanna Cheng** (JHU, co-advised with Homanga Bharadhwaj) 2026 – now
- **Jiafeng Gu** (JHU) 2026 – now
- **Rayhaneh Behravesh** (JHU) 2026 – now
- **Yixin (Tracy) Zhu** (UChicago); Project on Improving Projective Geometry 2024 – 2025
- **Feiran Wang** (UIUC); Project on NeRF + Latent Illumination 2023
- **Kexuan (Klaus) Zou** (UIUC | Now: Software Engineer at NVIDIA); Project on 2D Meshes 2019 – 2020

Undergraduate Students

- **Jin Hong (Jayden) Moon** (JHU) 2026 – now
- **Jiuzhou (Anthony) Cheng** (JHU) 2026 – now
- **Anudeep Chimata** (BITS Goa, India); Project on Representation Learning & Generative Models 2025 – now
- **Joshua Ahn** (UChicago); Project on Wavelets in Neural Radiance Fields 2024

- **Zhiyan (Alex) Wang** (UChicago); Project on Improving Projective Geometry 2024 – 2025
- **Kuan-Sheng Chen** (UIUC); Project on Improving Generative Models with Intrinsic Images 2023
- **Brian Chen** (UIUC | REU at CMU's RI | Next: Hive AI); Project on Lighting Correction 2020 – 2022
- **Anchu Zhu** (UIUC | Next: MS in CS at USC); Project on Anomaly Detection 2018 – 2019

Student Collaborators

Students whom I actively co-mentor or collaborate with (home institution in parentheses).

Doctoral-Level

- **Feng Wang** (JHU); **Papers in NeurIPS 2026** & under review 2026 – now
- **Namitha Guruprasad** (JHU) 2026 – now
- **Timing Yang** (JHU); **Papers in NeurIPS 2026** 2026 – now
- **Vivekjyoti Banerjee** (JHU); Paper under Review 2026 – now
- **Zhaoliang Zhang** (JHU) 2026 – now
- **Ozgur Kara** (UIUC) 2026 – now
- **David Serrano-Lozano** (UAB, Spain); **Paper in NeurIPS 2026** 2025 – now
- **Prakhar Kaushik** (JHU); **Paper in CVPR 2026** 2025 – 2026
- **Soumava Paul** (JHU); **Paper in CVPR 2026** 2025 – now
- **Tianchong Jiang** (TTIC); **Paper in ICRA 2026** (**Best Paper Award**) 2025 – 2026
- **Vinayak Gupta** (University of Maryland); **Paper in CVPR 2026** 2025 – 2026
- **Jun Myeong Choi** (UNC Chapel Hill); **Paper in CVPR 2025** 2024 – 2025
- **William Gao** (UChicago); **Paper in NeurIPS 2024** 2024 – now
- **Xiaoyan Xing** (University of Amsterdam); **Papers in CVPR 2025 & ICML 2026** 2024 – now
- **Ayush Sarkar** (UIUC); **Paper in CVPR 2024** 2023 – 2024
- **Kalyani Marathe** (University of Washington); **Paper in CVPR-W 2024** 2023 – 2024
- **Matthew Wallingford** (University of Washington); **Paper in NeurIPS 2024** 2023 – 2024
- **Vaibhav Vavilala** (UIUC); **Papers in ICLR 2026, 3DV 2026 & NeurIPS 2026** 2023 – 2025
- **Xiang Fan** (University of Washington); **Paper in ECCV 2024** 2023 – 2024
- **Xiao Zhang** (UChicago); **Papers in NeurIPS 2024 & ICML 2026** 2023 – 2025
- **Xiaodan Du** (TTIC); **Paper in CVPR-W 2024** 2023 – 2024
- **Zhi-Hao Lin** (UIUC); **Paper in CVPR-W 2024 & Paper in 3DV 2025** 2023 – 2024
- **Zitian Zhang** (Université Laval); **Papers in WACV 2025 & 3DV 2026** 2023 – 2024

Master's-Level

- **Naveen Chandra R** (IISc, India); **Paper in NeurIPS 2026** 2026 – now
- **Shivam Tripathi** (IISc, India); **Papers in CVPR 2026 & NeurIPS 2026** 2025 – now
- **Varun Varma Thozhiyoor** (IISc, India); **Papers in CVPR 2026 & NeurIPS 2026** 2025 – now
- **Frédéric Fortier-Chouinard** (Université Laval); **Papers in WACV 2025 & 3DV 2026** 2023 – 2025
- **Hanlin (Asher) Mai** (UIUC); **Paper in CVPR 2024** 2023 – 2025
- **James Soole** (UIUC | Now: Research Engineer at MATLAB); **Paper in CVPR 2024** 2023
- **Seemandhar Jain** (UIUC | Now: PhD at UCSD); **Papers in ICLR 2026, 3DV 2026 & NeurIPS 2024** 2023 – 2024
- **Pranav Asthana** (UIUC | Now: PhD at UMD); Project on Relighting from Neighbors 2021 – 2022

Undergraduate-Level

- **Furkan Horoz** (METU, Turkey / UIUC) 2026 – now
- **Pradhaan S Bhat** (PES University / IISc, India); **Paper in NeurIPS 2026** 2026 – now
- **Amitabh Mahapatra** (UIUC | Now: PhD at UIUC); **Paper in CVPR 2024** 2023 – 2025

- **Oscar Michel** (AI2 | Now: PhD at NYU); **Paper in NeurIPS 2023** 2023
- **Liwen Wu** (UIUC | Now: PhD at UCSD); **Paper in CVPR 2022 (Best Paper Finalist)** 2022

Doctoral Thesis Committees

- **William Gao**, "Neural Methods for 3D Mesh Editing" (UChicago) 2025
- **Xiao Zhang**, "Representation Learning from and for Generative Models" (UChicago) 2025
- **Xin Yuan**, "Interpretable Unsupervised Generative Learning via Factorized Architectures and Structured Bottlenecks" (UChicago) 2024

Courses Designed

Generative Vision: The Art and Science of Visual Synthesis, Johns Hopkins University; *Course Instructor* Fall 2026

- Developed a new course on visual synthesis with generative models

Learning-based Vision, Johns Hopkins University; *Course Instructor* Spring 2026

- Developed a new course on modern approaches in computer vision
- Taught a class of 35 students from Johns Hopkins University

Past Meets Present: A Tale of Two Visions, TTIC; *Course Instructor* Spring 2024

- Developed a new course from scratch on Computer Vision, combining historical and modern approaches
- Taught a class of 20+ students from TTIC and UChicago

Teaching Assistant

Computer Science, UIUC; *Graduate Teaching Assistant* 2016 – 2018

- Applied Machine Learning (CS498 AML), Fall 2018
- Data Structures (CS 225), Spring 2017
- Intro Computing: Engrg & Sci (CS 101), Spring 2016 & Fall 2016

Invited Talks

The Three P's of Modern Computer Vision: Pixels, Perception, and Physics

- University of Maryland, College Park, Vision Seminar Oct 2026
- Indian Institute of Science, Bangalore, Kotak-IISC AI/ML Seminar Jan 2026
- Johns Hopkins University, Guest Lecture in Rama Chellappa's Machine Perception Course Nov 2025
- Johns Hopkins University, Physics of Learning Seminar, invited by Brice Ménard Oct 2025
- Johns Hopkins University, ECE Department Seminar Sep 2025

Principia: What Video Models Know about High-School Physics

- CVPR AC Workshop (Lightning Talk) Jun 2026

Emergent Latent Intrinsic Representations for Scene Relighting

- UChicago, Great Lakes Graphics Workshop Apr 2025

What Generative Image Models Understand (and Don't) about the Physical World

- Univ of Wisconsin at Madison Apr 2025
- Johns Hopkins University, CS/LCSR Seminar Mar 2025
- UC Irvine, CS Seminar Mar 2025
- Stony Brook University, CS Seminar Mar 2025
- UIUC: External Speaker Series Feb 2025
- ULaval: Vision Seminar Feb 2025
- UCSD: Pixel Cafe Seminar Jan 2025

Are Generative Image Models Physically Grounded?
- UPenn: GRASP Seminar — Dec 2024
- NYU: Guest Lecture in Saining Xie's Computer Vision Course — Dec 2024

Generative Models Inside Out
- Midwest Computer Vision Workshop, Indiana University — Sep 2024

What Do Generative Image Models Know?
- IIIT Hyderabad, India; Vision Seminar — Jan 2024
- Exactech, Inc.; Tech Talk — Oct 2023

Exploring Knowledge in Generative Models
- Stanford University in Jiajun Wu's group — Jun 2023
- University of Tubingen, Autonomous Vision Group — May 2023

What Do Generative Models Know about Geometry and Illumination?
- UC Berkeley: Vision Seminar — Apr 2023
- NVIDIA Research — Apr 2023
- MIT: Vision and Graphics Seminar — Apr 2023
- CMU: VASC Seminar — Mar 2023
- UW: RAIVN Vision Seminar — Mar 2023

Towards a Productive and Fun PhD Experience
- UIUC: Computer Vision Workshop — Apr 2023

Learning about Light Without Labeled Data
- UMD: Vision Seminar — Mar 2023
- UCSD: Pixel Cafe Seminar — Feb 2023
- TTIC: Research Talk — Feb 2023

Drag-and-Drop Rendering: Towards In the Wild Image Editing
- CMU: Misc-Read Vision Reading Group — Aug 2021

Services & Leadership

Reviewing
- **Area Chair**, ICLR 2027, CVPR 2027, NeurIPS 2026, CVPR 2026, ICLR 2026, 3DV 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, WACV 2025 — 2024 – now
- **Reviewer**, CVPR, NeurIPS, ECCV, ICCV, ICLR, SIGGRAPH, SIGGRAPH Asia and PAMI — 2018 – now

University Service
- **Faculty Hiring Committee**, Department of Computer Science, Johns Hopkins University — 2025 – 2026

Community Building Workshops
- **Lead Organizer**, "Bitter Lessons" workshop at CVPR — Jun 2026
- **Lead Organizer**, "How to Stand Out in the Crowd?" workshop at CVPR — Jun 2025
- **Lead Organizer**, "CV 20/20: A Retrospective Vision" workshop at CVPR — Jun 2024
- **Lead Organizer**, "Scholars & Big Models: How Can Academics Adapt?" workshop at CVPR — Jun 2023

Technical Workshops
- **Organizer**, Findings, a new track at CVPR — Jun 2026
- **Organizer**, "Humans of Generative Models (HuGs)" at CVPR — Jun 2026

- **Lead Organizer**, “UniLight Workshop” at ICCV Oct 2025
- **Co-Organizer**, “Ind3D: Enforcing geometric, physical, topological, and functional inductive bias in 3D generation” workshop at CVPR Jun 2025
- **Lead Organizer**, “Knowledge in Generative Models” workshop at ECCV Sep 2024
- **Co-lead Organizer**, “Multimodal Artificial Intelligence ” workshop at TTIC Aug 2024
- **Lead Organizer**, Computer Vision workshop at Allerton, UIUC Apr 2023
- **Committee**, “Adversarial ML in Real-World Computer Vision Systems", workshop at CVPR Jun 2019
- **Committee**, “Security and Privacy in Machine Learning" workshop at ICML Jun 2019

Seminars and Reading Groups

- **Organizer**, UIUC Vision Seminar, a new weekly seminars inviting speakers outside UIUC 2021 – 2023
- **Organizer**, Vision Lunch (computer vision reading group) 2021 – 2023
- **Organizer**, UIUC Summer Vision Coffee 2018

Co-Founder, SchoolEngg 2013 – 2015

- Forum for high-school students; providing firsthand knowledge of all engineering disciplines
- Renamed as *PrepLift*, an education counselling start-up in India

Co-Founder, American Society of Civil Engineers (ASCE) NITK Chapter 2013 – 2015

- Co-Founded the *first* ASCE student chapter of India
- Served as treasurer for one year and as a senior advisor for the second year
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events